Module 9: Ethics and Responsibility

Accountability, transparency, and team practice

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Primary sources: Osmani, Chapter 9; Thomas & Hunt, Topic 49: Pragmatic Teams.

Learning Outcomes

- Evaluate ethical risks in AI-assisted development.
- Assess responsibility boundaries between engineers and AI systems.
- Analyze bias and unintended consequences.
- Justify responsible engineering practices.

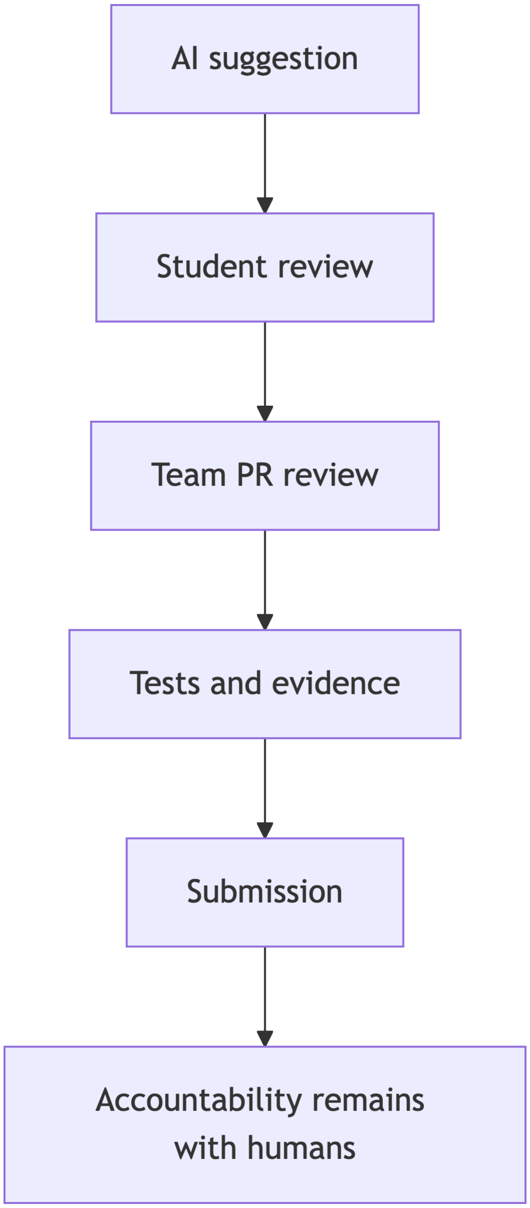
Responsibility Boundary

AI can assist, but it cannot be accountable.

The engineer or team remains responsible for:

- Submitted code.
- Security impacts.
- Documentation accuracy.
- Disclosure of AI assistance.
- User consequences.

Accountability Flow



Transparency

Responsible AI use includes a clear artifact trail.

- What tool or mode was used?
- What prompt or task was given?
- What was accepted, modified, or rejected?
- How was the result verified?
- What uncertainty remains?

IP and Provenance

AI-generated output can raise provenance questions.

Practical safeguards:

- Avoid asking for copyrighted code.
- Be cautious with suspiciously specific output.
- Prefer explanations, patterns, and tests over copied solutions.
- Run license or provenance checks where required.
- Follow project policy before accepting generated code into deliverables.
- Follow course and organizational policy.

Privacy and Consent

Responsible AI use also means protecting inputs.

- Do not paste secrets, private user data, or proprietary code into tools without permission.
- Use sanitized or synthetic examples when debugging.
- Check tool settings and organizational policy for data retention or training use.
- Disclose AI-backed user-facing features when users need to know.
- Provide notice, consent, opt-out, or deletion controls where required by law, policy, or user expectations.

Bias and Consequences

Software can encode unfair assumptions.

AI can amplify them by producing common patterns without context.

Ask:

- Who could be harmed?
- Whose data is missing?
- What assumptions are hidden?
- What feedback would reveal harm?

Responsible AI Checklist

Osmani's responsible-use guidance turns ethics into reviewable practice.

- Keep a human in the loop.
- Take responsibility for code and decisions.
- Comply with relevant laws, licenses, and policies.
- Create guardrails and safety nets.
- Document AI usage decisions.
- Test for bias, discrimination, and misuse.

Pragmatic Teams

Thomas and Hunt emphasize stable, communicative teams with shared standards.

For AI-assisted teams:

- Agree on AI usage rules.
- Review AI-generated code carefully.
- Share prompt patterns that work.
- Do not let AI create invisible solo work.

Pragmatic teams make responsibility collective: quality is built in by everyone, not delegated to the person who clicked accept.

Team Norms for AI

Borrowing from Pragmatic Teams, responsible AI work needs team-level habits.

- Maintain awareness so scope, risk, and AI use do not drift unnoticed.
- Keep communication low-friction when prompts or outputs surprise you.
- Use end-to-end slices so all disciplines see real consequences early.
- Automate checks where possible instead of relying on memory.
- Schedule process reflection as AI tools and policies change.

Controlled Collaboration

Team repositories should show:

- Issues or task board.
- Feature branches.
- Pull requests.
- Review comments.
- Test evidence.
- AI usage notes.

Lab 5 Development Connection

Coding agents increase ethical stakes because they act over multiple steps.

Controlled workflows need:

- Human checkpoints.
- Reproducible runs.
- Tool boundaries.
- Reviewable artifacts.
- Clear stop conditions.

Practice Activity

Draft team AI rules:

- When is Copilot allowed?
- What must be documented?
- What requires peer review?
- What data must never be pasted?
- What counts as sufficient verification?

Takeaways

- Responsible use is an engineering practice, not a slogan.
- Transparency supports grading, collaboration, and trust.
- Teams need explicit norms for AI-assisted work.